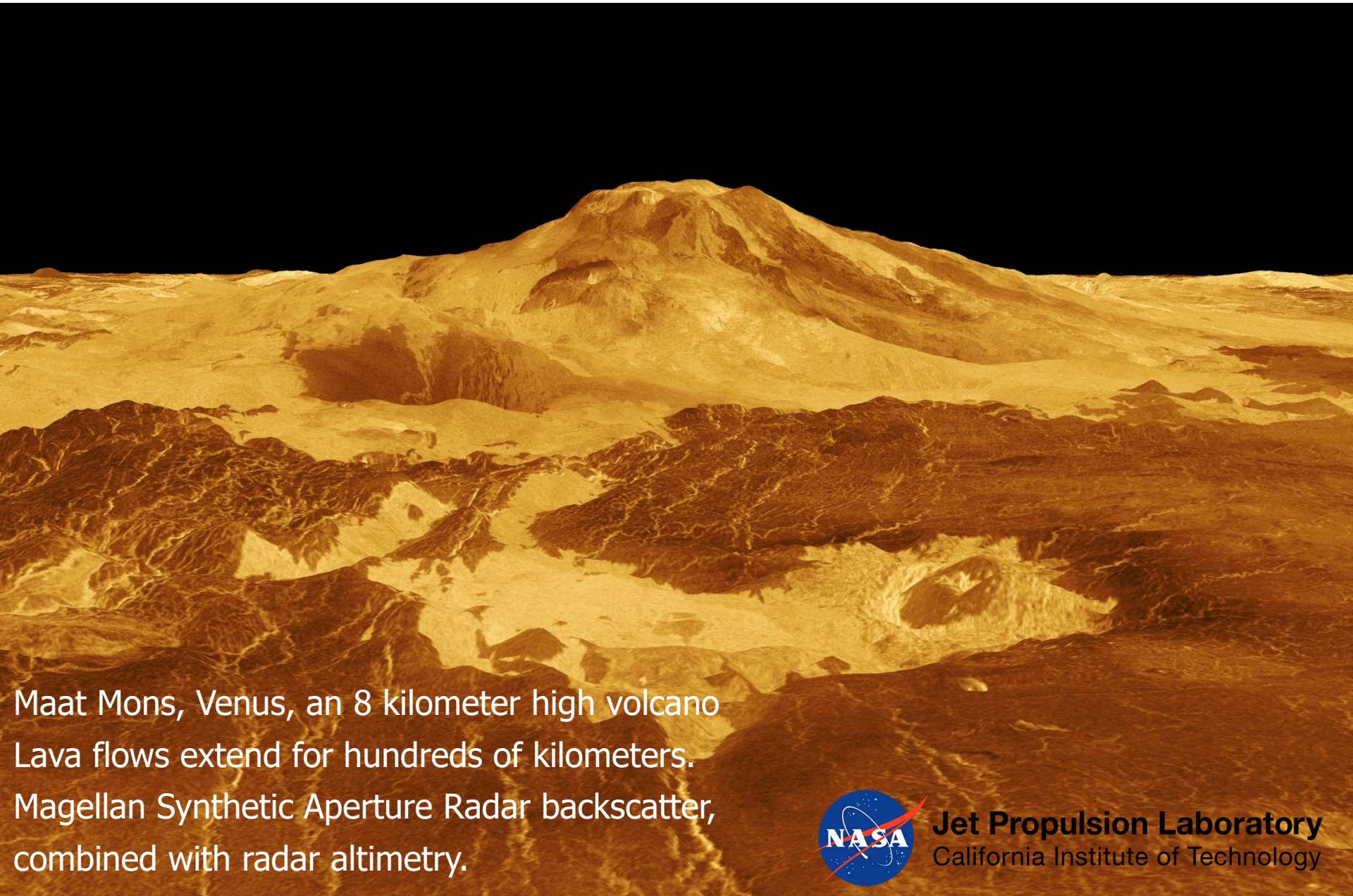
RAdio Detection And Ranging



Maat Mons, Venus, an 8 kilometer high volcano
Lava flows extend for hundreds of kilometers.
Magellan Synthetic Aperture Radar backscatter,
combined with radar altimetry.

Jet Propulsion Laboratory
California Institute of Technology

In this lesson:

- Monodimensional Radar: components
- Mathematical modelling of radar acquisition and signal processing
- Estimation of target **amplitude, range (position), phase**
  for single – isolated targets, and single pulse
- The role of SNR
- An industrial application example

Radar components



Radar equipment



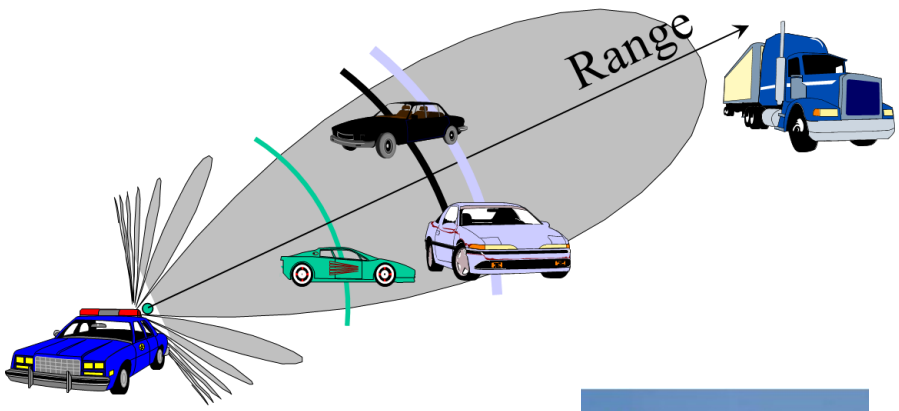
Radar Scene and targets

Mathematical models & digital signal processing

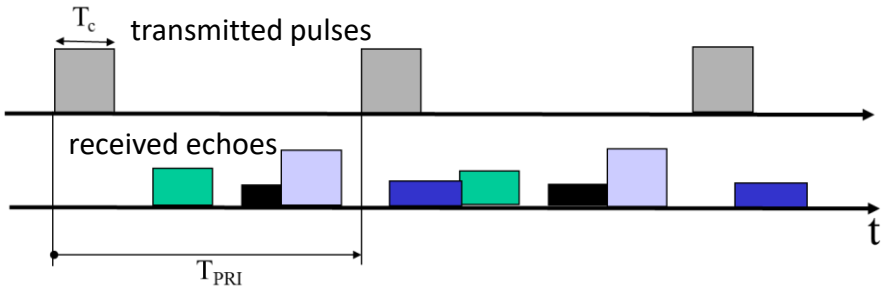
Signal processing and results



Pulsed Radar: overview

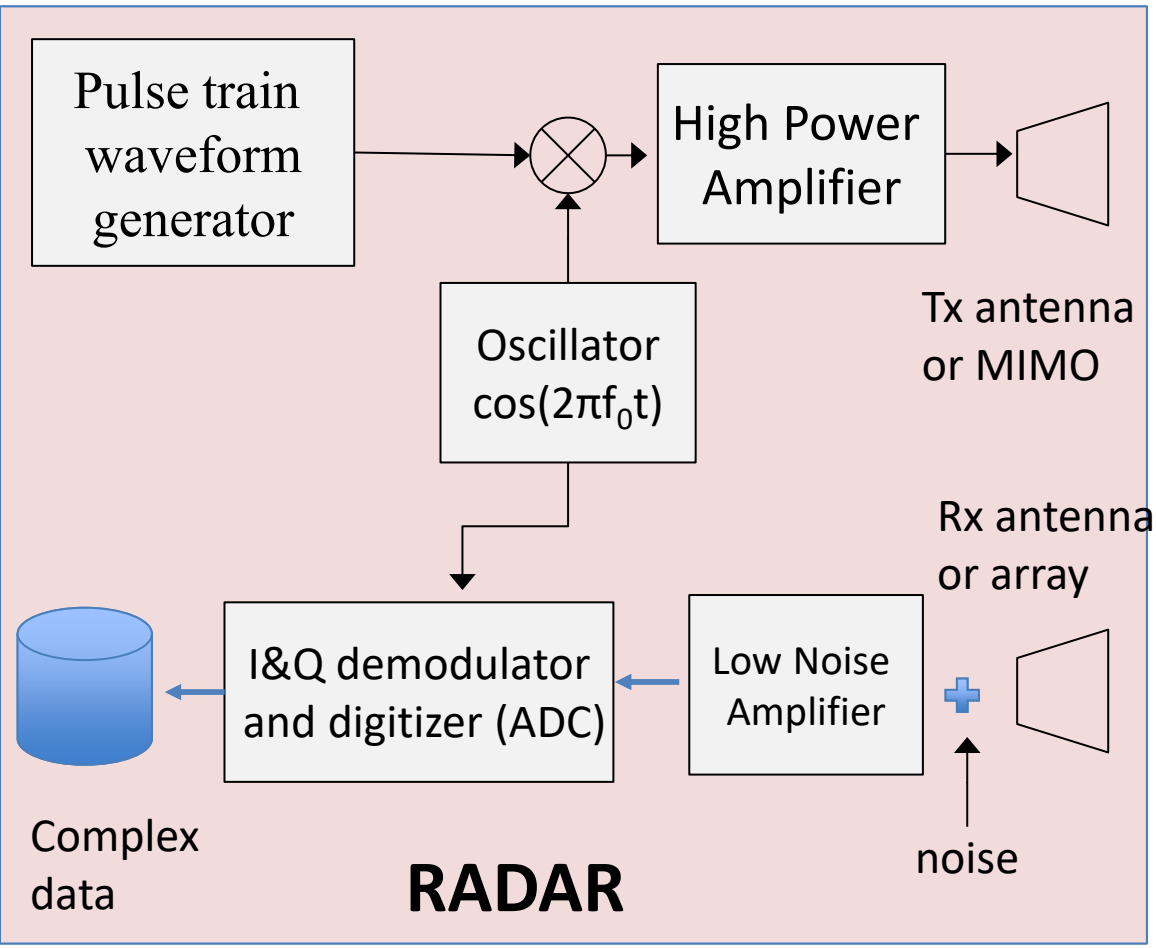


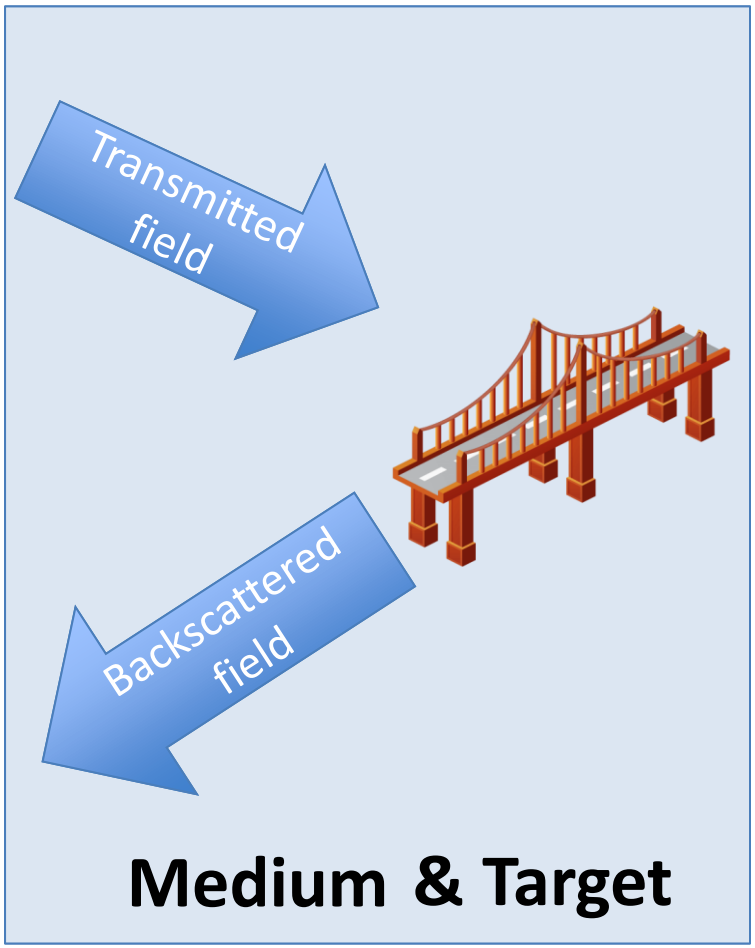




- We use the polar coordinates, with its origin **at the Radar position**.
- A target in $P(r_P)$, produces a backscattered return with **amplitude** ρ_P, and **pha**se ϕ_P
- Its 2D position can be detected by (1) the range and (2) the antenna look direction
- A resolution cell may contain zero, one, or several targets
- The system is linear (superposition of effects)
- The echo model is linear: contributions from individual targets superpose
- The waveform is a modulated pulse train

Pulsed Radar: overall block diagram





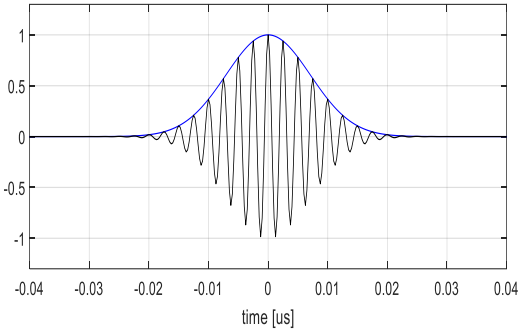
Radar is a telecommunication system. We know the transmitted and received signal and we infer properties on target.

Model analysis

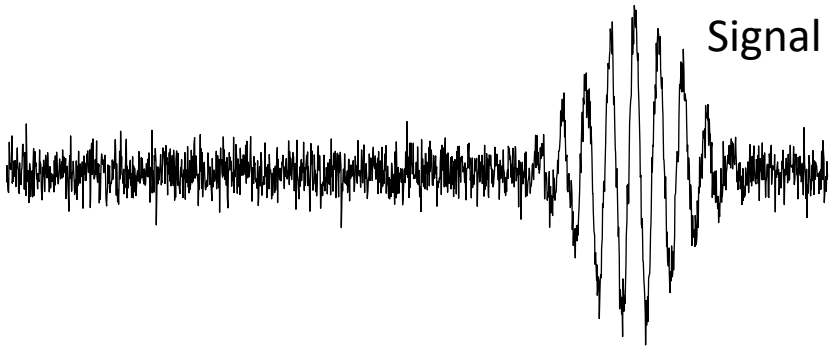
The purpose is to evaluate:

1) the amplitude of the target
2) the phase of the target
2) the position of the target (range)
3) the velocity of the target
   (from a sequence of pulses)

Transmitted pulse $s_T(t)$



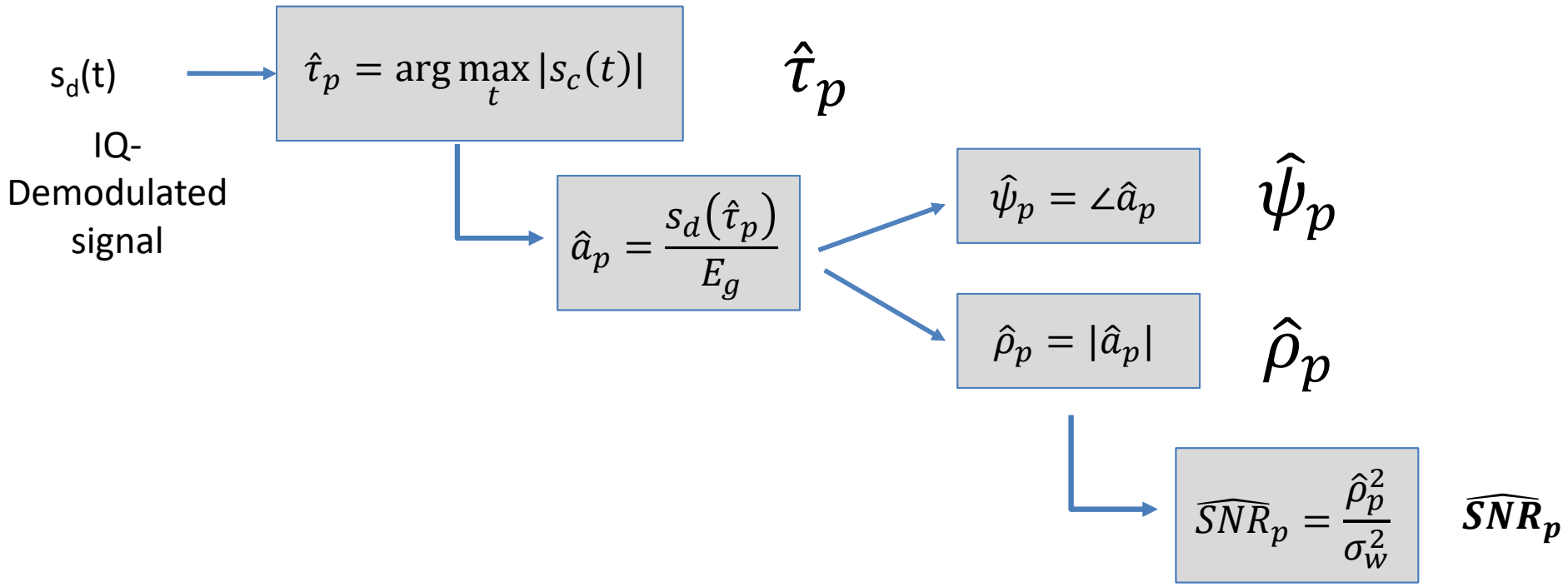
Model of the signal before / after the matched filter



Signal model before demodulation & matched filtering

$$s_c(t) = \rho_p \cdot e^{j\psi_p} \cdot r_g(t - \tau_P) + w_c(t) * g^*(-t)$$

Estimation of target range, amplitude & phase



Application: Infrastructure Health Monitoring

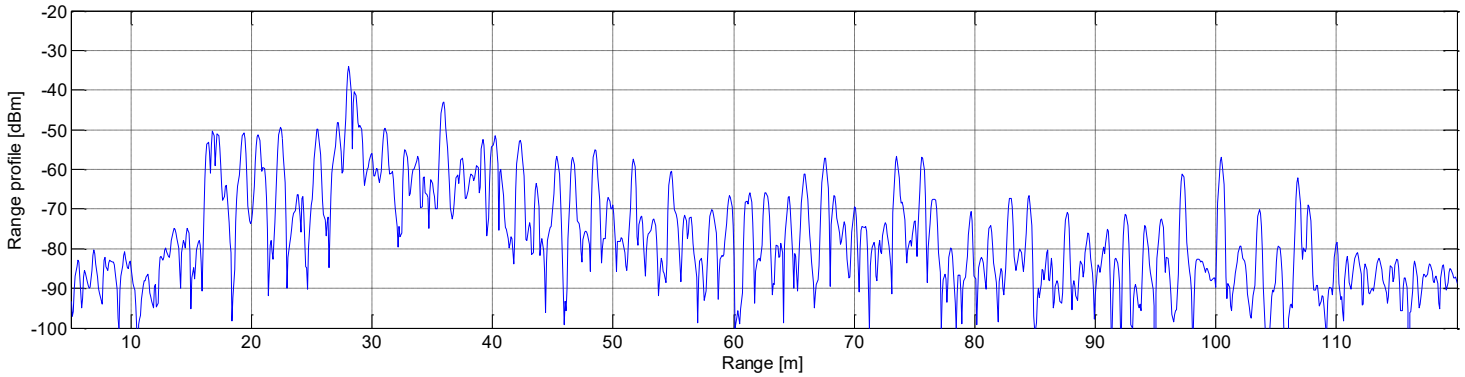
OBJECTIVE: measure bridge displacement and vibration during a train crossing.

Bridge Under test

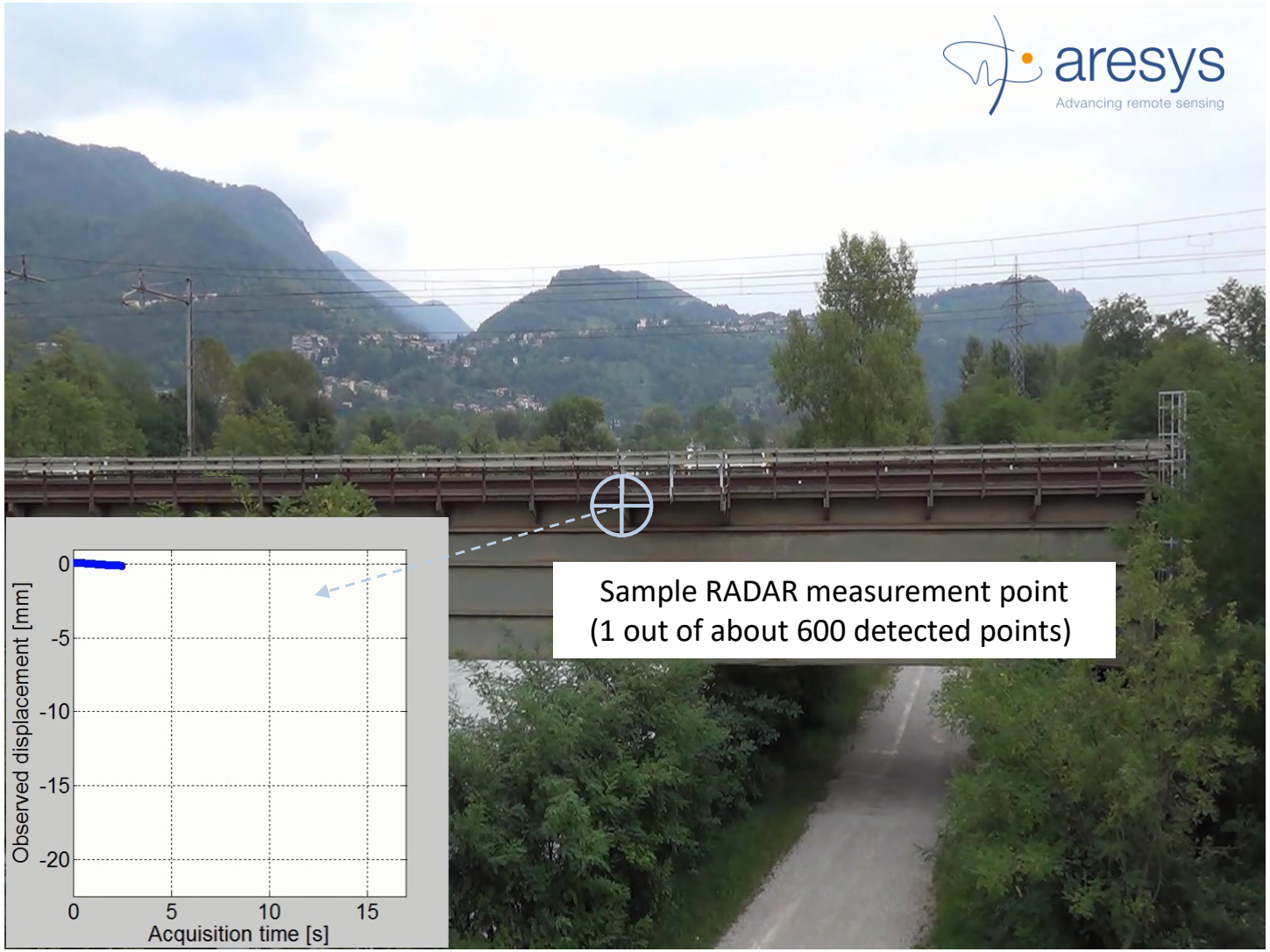




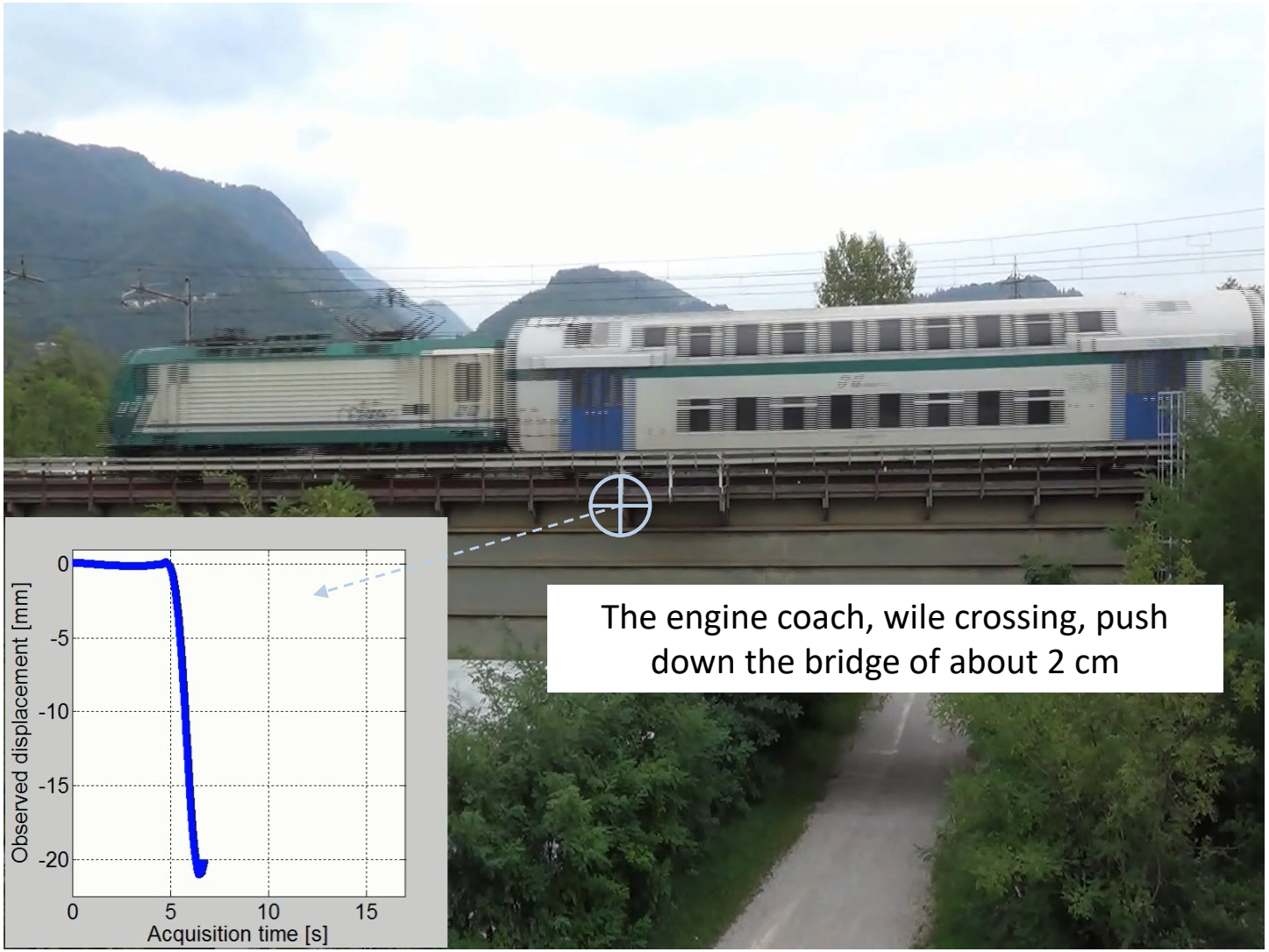
Range progfile power



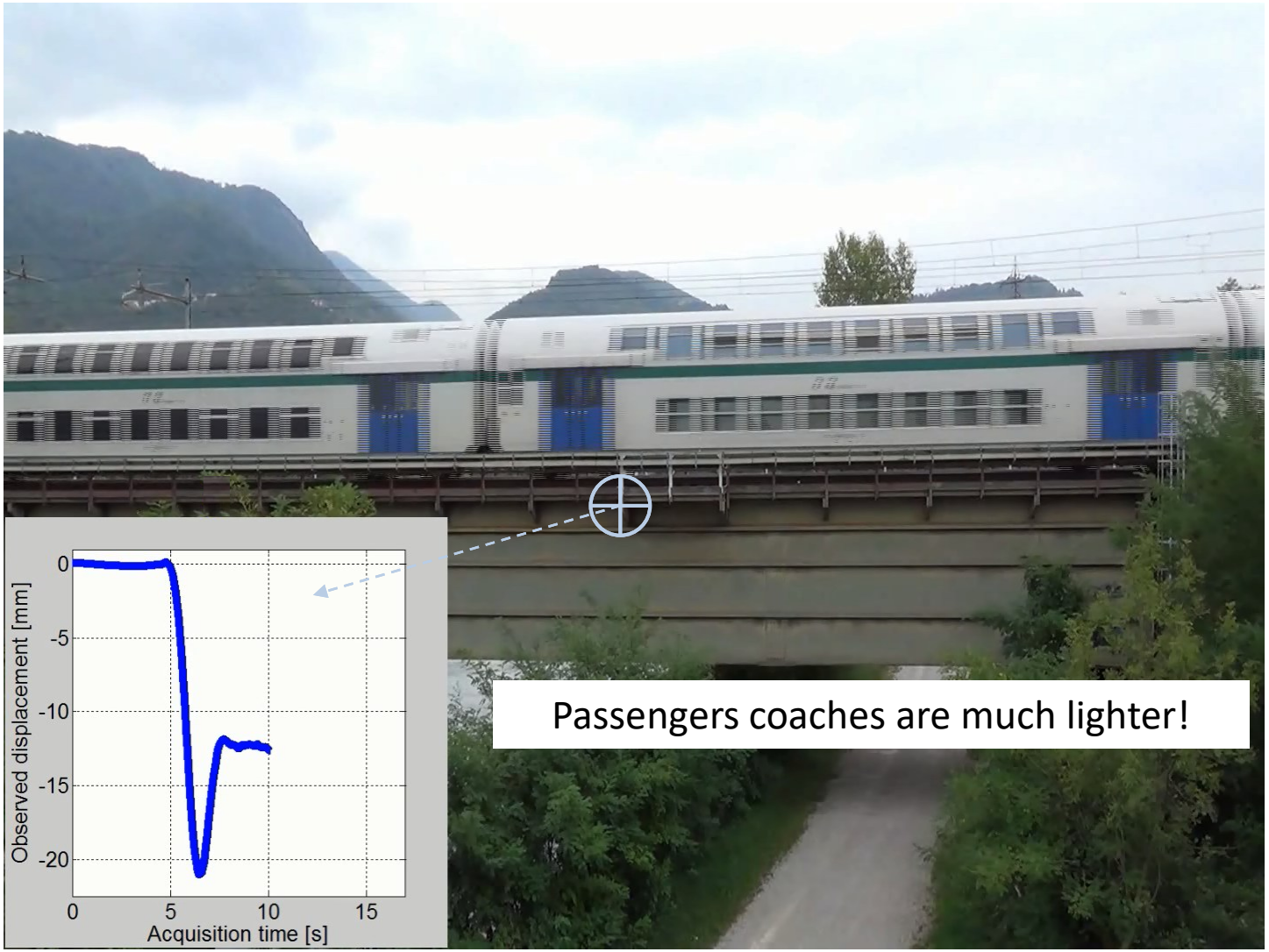
Application: Infrastructure Health Monitoring



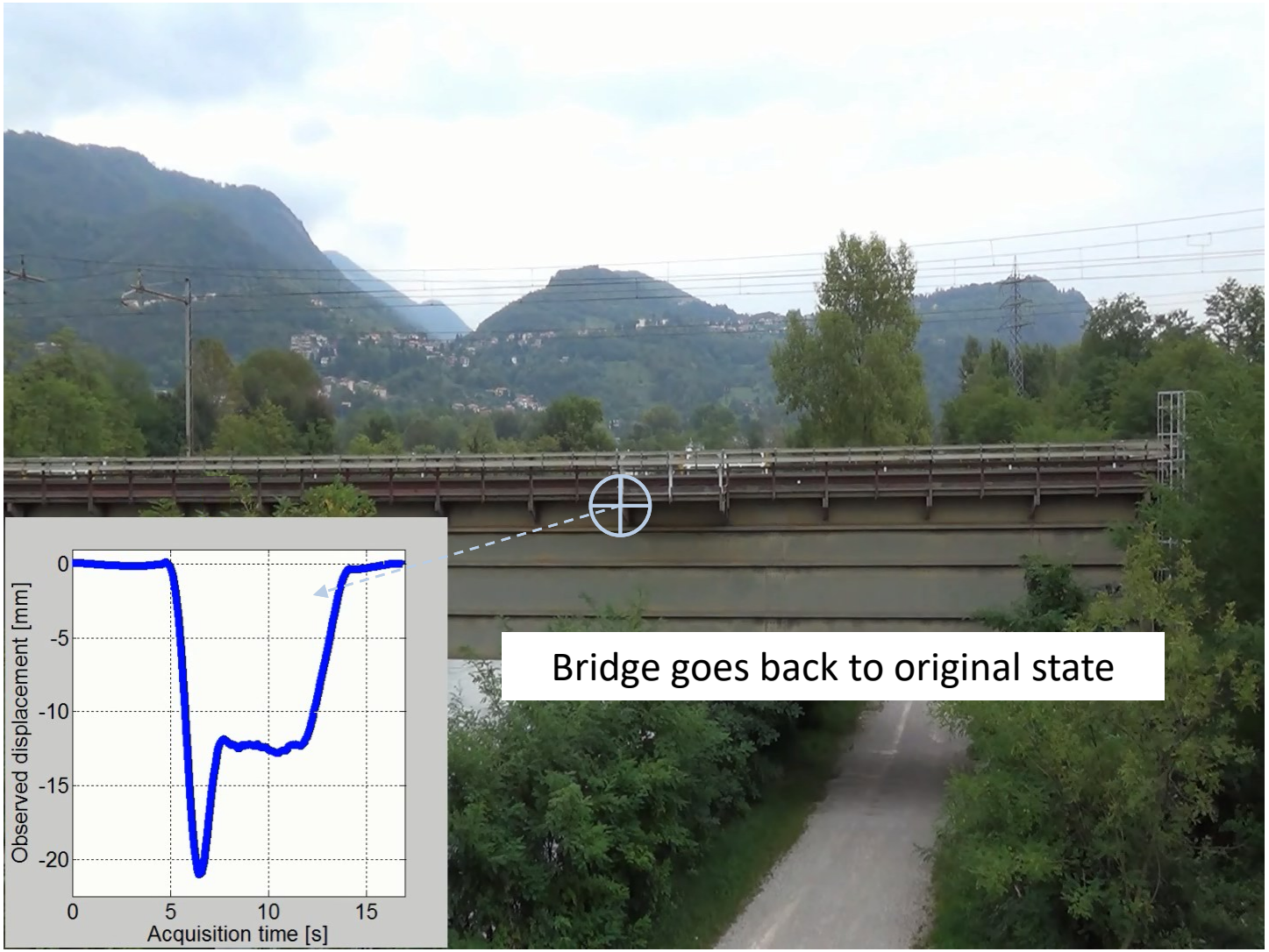
Application: Infrastructure Health Monitoring



Application: Infrastructure Health Monitoring

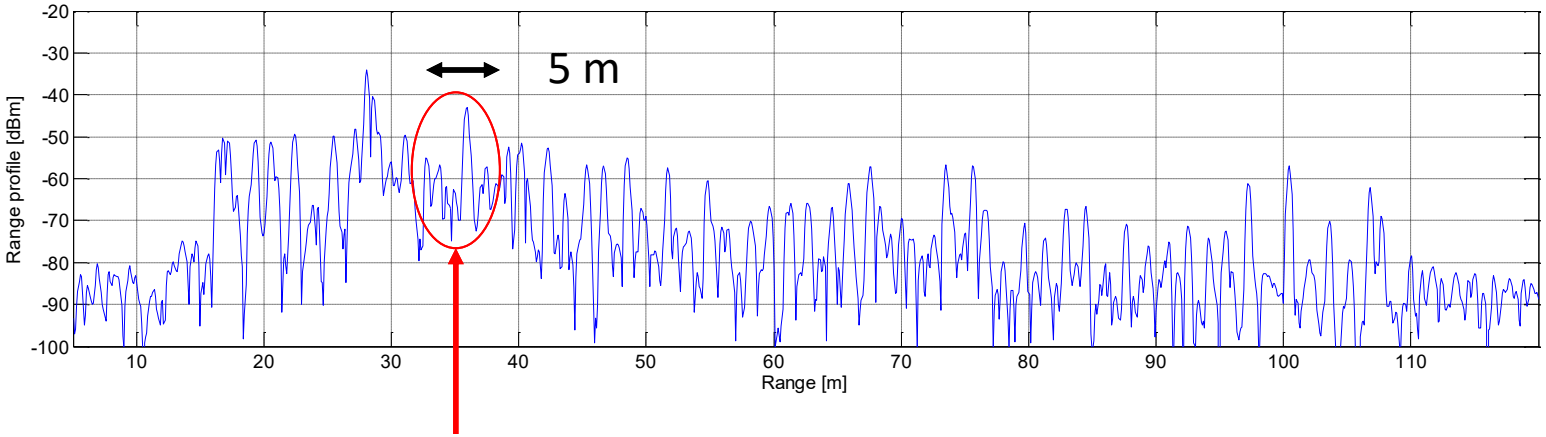


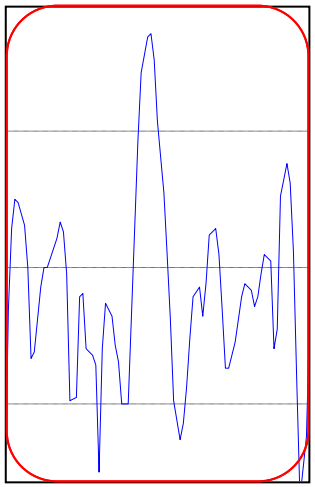
Application: Infrastructure Health Monitoring



Estimating displacement and velocity

Backscattered power versus range





Target of interest r_p = 35 m

If the target moves by a distance much smaller than the range resolution, can its displacement still be measured?